

From 2.5D to 3D Chiplet Systems: Investigation of Thermal Implications with HotSpot 7.0

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Abstract—Recent advanced packaging technologies such as 2.5D chiplet-package offer a modular approach to increasing yield over monolithic SoC designs. As 2.5D chiplet systems shed light on reducing product development times and costs, 3D chiplet systems can extend the benefits furthermore by offering more remarkable performance. As semiconductor technology continues, the significance of thermal management has arisen even for a monolithic chip. When it comes to 3D chiplet systems, thermal issue remains being one of the most critical obstacles to transitioning from 2.5D to 3D. Microfluidic cooling has been proved to be a promising cooling solution, yet the actual detailed thermal indications for making the direct transition under this cooling strategy are still missing. HotSpot 7.0 emerged as the latest pre-RTL thermal analysis framework that introduced a novel thermal management method using microfluidic cooling on the widely used pre-RTL power and thermal simulator. This paper presents evolution investigations from 2.5D to microfluidic-cooled 3D integration from the thermal management aspect using HotSpot 7.0. We studied a typical 2.5D chiplet as an example, and it has one processor chip in the center and four high bandwidth memory (HBM) chips on the sides. The thermal management method with microfluidics reduces the maximum temperature of 2.5D and 3D chiplet by 47.2°C and 63.83°C, respectively. In addition, as a high-performance system example, a hypothetical processor-processor integration is investigated. The simulation results show that the conventional air convection type is hard to maintain the chiplet temperature under the operating temperature range. Microfluidic cooling is advantageous in heat dissipation and heat spreading. The cooling capacity is dependent on the pump pressure. Multi-layer cooling is a promising cost-effective solution for the 3D chiplet system.

Index Terms—HotSpot, chiplet, thermal management, thermal simulation, microfluidic cooling

I. INTRODUCTION

As the semiconductor technology scaling continues, the associated manufacturing costs have also increased steadily. This increase has been even sharper in recent generations due

to increased mask costs, extreme ultraviolet lithography (EUV) costs, more complex fabrication processes, and more. But from the application perspective, the need for higher performance, more functionality and on-chip storage, and lower cost has never been so demanding. These natural controversies led to advanced packaging solutions such as chiplets, shifting the design focus to system level. A chiplet is a sub-processing unit, usually controlled by an I/O controller chip on the same package. The silicon interposer-based chiplet system is not dramatically different from what we have today for an SoC-style design, but at a larger scale. Each chiplet can be manufactured using a separate process, and the routing can be planned as if it was for one die. A chiplet system is able to increase the yield since a single monolithic system on a chip with a defect will need to be either thrown away or sold as a lower model. Chiplet solution provides the opportunity of a single defective core and assembling good dies so it can be sold as the desired model. This has dramatically offered leverage of cost/yield vs. computing demands [1]–[4].

Current cutting-edge chiplet systems such as 2.5D integration or chip-on-wafer solutions achieved better performance metrics with reduced power consumption. For the 2.5D integration, multiple dies in a single package are located side by side on a silicon interposer, an interconnection layer. The interposer acts as the bridge between the chips and a board, which provides more bandwidth and I/Os. An Example of such an interconnect is the embedded multi-die interconnect bridge (EMIB) from Intel to enable die-to-die integration on package [5]. While chiplet-based systems offer more significant advantages over the monolithic chip approach, it is mainly limited to a 2D dimension with limited 3D extension but primarily for memories. A well-known example is high-bandwidth memories (HBMs) – stacks of dynamic random access memory (DRAM) chips [2]. This memory stack connects to other chiplets through interface buses, limiting their use to latency-tolerant applications. Consequently, the 2.5D chiplet-based

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system will never allow for fast access between logic and fast, first and intermediate level cache memories, or even between logic [1]. Thus, from the interconnection aspect, 3D IC is even more advantageous in performance and power-saving because it significantly reduces the interconnection distance compared to the interposer.

For the 3D integration, multiple dies are stacked as a single chiplet utilizing through-silicon-vias (TSVs). An actual 3D IC chiplet system enables an advanced package to stack logic on memory or logic on logic. A number of recent 3D integration demonstrations have been explored to allow for opportunities in high-performance computing [6], image sensors [7], or gas sensing [8]. These are also known as monolithic 3D integration. However, one of the most critical factors that obstruct the 3D IC development is the thermal issue resulting from the increased power density. Power density is already an issue for high-frequency 2D monolithic chips in advanced technology nodes [1]. The temperature of 3D IC rises over operating temperature because of the dimension mismatch between heating and cooling. While the 3D IC generates volumetric heat with increased power dissipation, the top surface dimensions limit the conventional cooling capacity. This thermal problem will get progressively worse as more computing-intensive units (such as AI computing units or GPUs) stack on top of each other or a CPU [4], [9]–[11].

Researchers have been actively looking into novel cooling solutions for dealing with 3DIC thermal issues [9]–[11]. But there is still no practical solution that can fully resolve the thermal issues in 3DIC. Recent studies have investigated microfluidic cooling as one of the most promising thermal management solutions for future 3D IC chiplet systems [12]–[15].

While for chip designers or architects, there is an emerging demand to connect the dots between design and cooling during the early phase, even before the actual design phase. For example, given a list of chiplets, what would be a good stacking solution from a thermal perspective, assuming microfluidic cooling is employed. Vice versa, given a known stacking solution, what would be an excellent microfluidic cooling solution to apply. By exploring the design space, chip designers can make early design decisions, and the design cycle can thus be shortened. This is critical, especially for 3DIC, as thermal management is a key factor for stacking strategies. Therefore, a fast and accurate simulation framework that integrates the most advanced microfluidic cooling modeling is necessary. We introduce HotSpot 7.0, a pre-RTL simulation framework that offers microfluidic thermal management design and simulation abilities in the early design step.

HotSpot 7.0 is the latest pre-RTL power and thermal simulator, built on top of the widely used HotSpot simulation framework [16]–[18]. This recent major update introduced the microfluidic cooling feature for 3D integrated circuits (IC). The distinctive advantage of this simulator is the design flexibility in thermal management tailored for 3D IC. It included 3D IC simulation features using 3D thermal circuit modeling consistent with the previous versions, and it adds

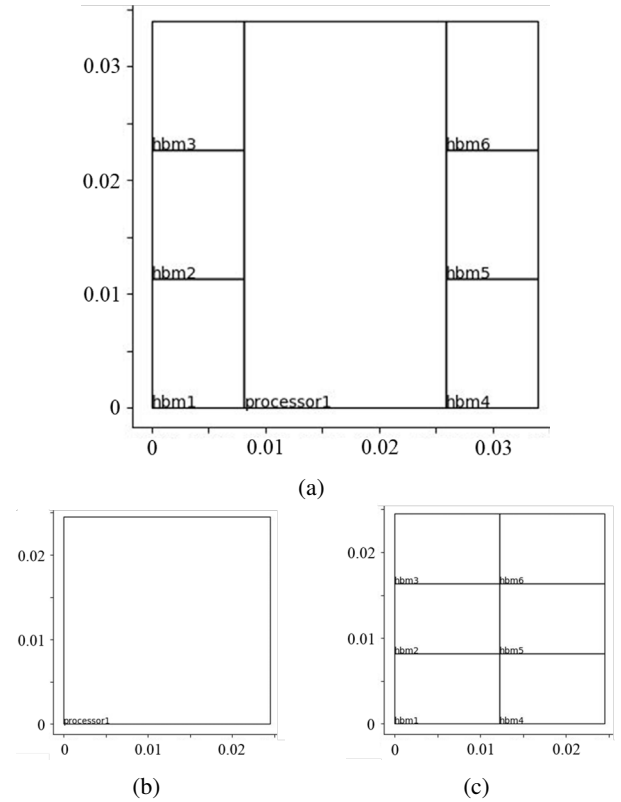


Fig. 1: Floorplan diagrams of (a) 2.5D chiplet integrating processor and memories and (b) processor layer and (c) memory layer of 3D chiplet. Vertical and Horizontal axis represents position of the chiplet in meters.

fluidic circuit modeling and pressure circuit modeling. With additional model layers, HotSpot 7.0 offers design flexibility exploration of microfluidic thermal management channel designs with different shapes, sizes, numbers, and flow rates.

In this paper, we present investigations of evolution from 2.5D to microfluidic-cooled 3D integration from the thermal management aspect using HotSpot 7.0. The contributions of the paper are summarized in the following.

II. THERMAL MODELING OF CHIPLET

We model the 2.5D chiplet system with two cases. The first case is a typical 2.5D chiplet example, and it has one processor chip in the center and six high bandwidth memory (HBM) chips on the sides. The 3D IC counterpart is configurable by stacking the memory chips on the processor. The size of the processor and memory chips are 552mm^2 and 92mm^2 , respectively. The floorplans of the chiplet in Figure 1 is generated using ArchFP [18]. We also create hypothetical power consumption as 100W for the processor and 18.4 W for the HBM. The operating temperature has been simulated with a passive heat sink, a low-end active heat sink, a high-end active heat sink, and a server-level heat sink. The convection thermal resistance is 4.0 K/W, 2.0 K/W, 0.5 K/W and 0.2 K/W, respectively [19]. It is simulated with microfluidic cooling

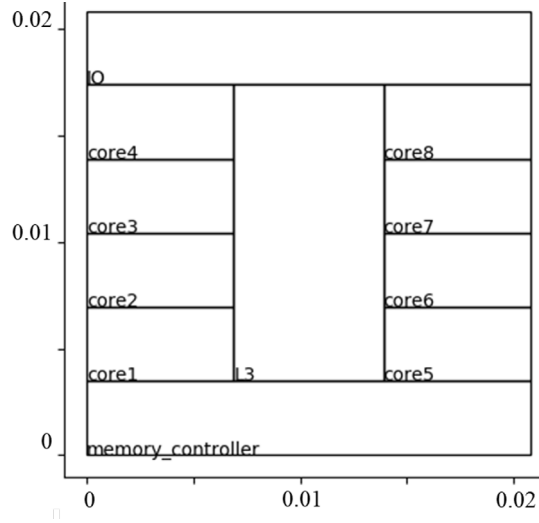


Fig. 2: Floorplan diagram of hypothetical processor

as well as conventional heat sinks. We assumed that the microchannel had integrated into the interposer layer. The 2.5D chiplet has 32 microchannels of a height of 100 μm and a width of 346 μm . And the 3D chiplet has 62 microchannels of 100 μm height and a width of 191 μm . The microchannel cooling features of 2.5D and 3D chiplet have a flow rate of 35.5 ml/min and 36.3 ml/min under the 200 mbar pump supply pressure, respectively. 200 mbar is affordable fluidic pressure to the 9V reservoir pumps of commercialized liquid cooling systems.

Next, we model a processor-processor integration which is a thermally severe example. The hypothetical processor architecture is modeled based on the Intel i7-3960X [20], [21]. Figure 2 shows the floorplan of still square with a side length of 20.8 millimeters generated with ArchFP. Each processor has integrated side by side for the 2.5D chiplet integration, and for the 3D chiplet, one processor is stacked directly on the other processor. We also created a hypothetical power map based on the i7-3960X's published TDP of 130 watts. The power-thermal characteristics are investigated under the heat sinks with convection thermal resistance of 4.0, 2.0, 0.5, and 0.2 K/W. And we use the same two configurations for the microfluidic cooling layer with 31 microchannels of a height of 100 μm and a width of 325 μm . The microchannels have a flow rate of 53.5 ml/min under a 200 mbar pump supply pressure.

III. RESULTS

To investigate the thermal implications of the transition from 2.5D to 3D chiplet, we perform thermal simulations using HotSpot 7.0. By exploring different cooling features, we study allowed maximum power densities and operating temperatures in 2.5D and 3D chiplet. In our first example, we model a chiplet composed of a processor and 3D-stacked memories. Figure 3 and 4 show the maximum temperature of the 2.5D and 3D chiplet with increasing power consumption

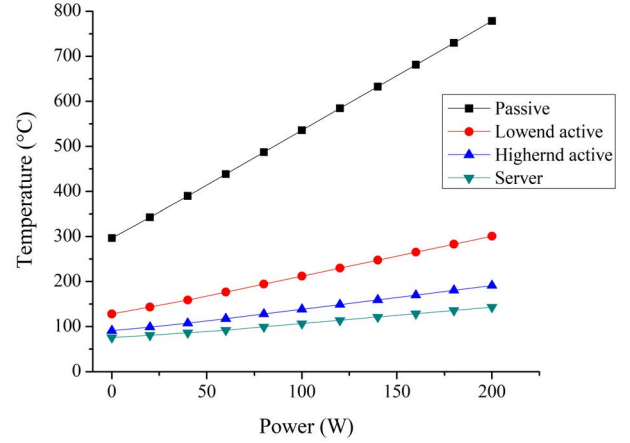


Fig. 3: Temperature of 2.5D chiplet with heat sinks.

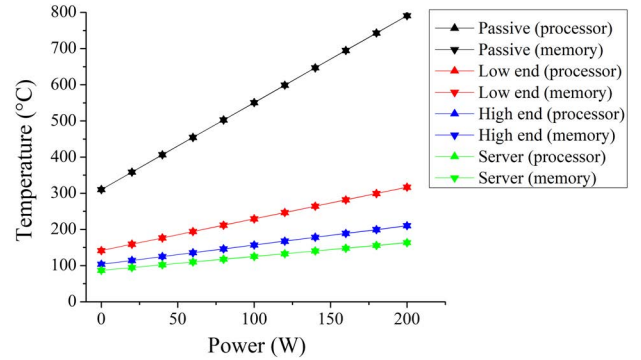


Fig. 4: Temperature of 3D chiplet with heat sinks.

in the processor. The power consumption of the memories remains at a constant 18.4 W. The passive and low-end active heat sinks cannot remove the heat flux generated from the 3D stacked memory alone. When the power consumption of the processor is 100W, the maximum temperature of the 2.5D chiplet with high-end and server heat sinks are 212.48°C and 138.66°C, respectively. The maximum temperatures of 3D chiplet at the same condition are 229.09°C and 156.81°C. The temperature of the 3D chiplet is higher than 2.5D due to the higher power density, but in either case of chiplet, the cooling capacity by the air-cooling heat sinks has reached the limit. Therefore, novel thermal management methods such as microfluidic cooling are required to maintain the operating temperature under the limit.

Figure 5 and 6 show the temperature comparison of 2.5D chiplet with convection thermal resistance is 0.2 K/W and microfluidic cooling. The maximum temperature of chiplet with microfluidic cooling is around 9.5°C lower than the heat sink. Microfluidic cooling has superior heat spreading characteristics as well as higher heat removal capacity. Although microfluidic is advantageous to heat sinks in cooling behavior, it still lacks the cooling capacity to maintain the operating

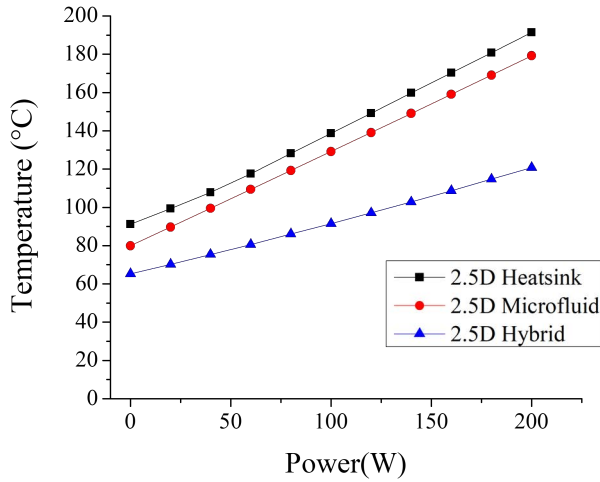


Fig. 5: Temperature comparison of 2.5D chiplet with server heat sink, microfluidic cooling, and hybrid of heat sink and microfluidic cooling.

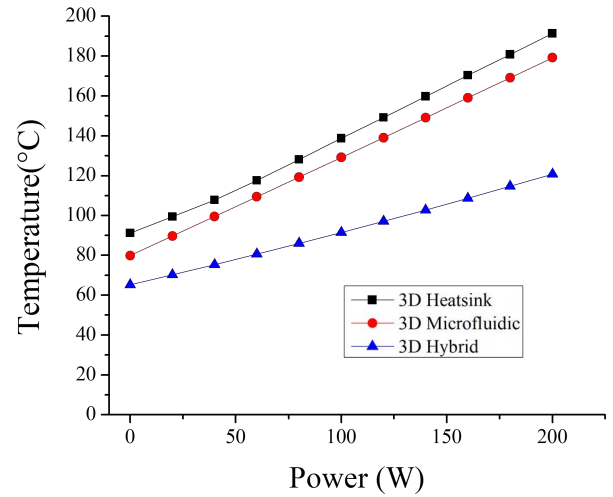


Fig. 7: Temperature comparison of 3D chiplet with server heat sink, microfluidic cooling, and hybrid of heat sink and microfluidic cooling.

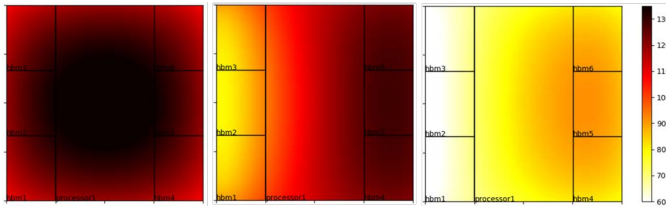


Fig. 6: Temperatures throughout a 2.5D chiplet with (a) server heat sink, (b) microfluidic cooling, and (c) hybrid of heat sink and microfluidic cooling at power dissipation of 100 watt.

temperature under the limit. Increasing pump pressure could be a solution, but adding another cooling layer is more effective. When the microfluidic cooling is integrated at interposer with the top surface heat sink, the operating temperature decreases 47.2°C.

Next, the 3D chiplet simulation results are shown in figures 7 and 8. Microfluidic cooling alone and with the heat sink decreases the maximum temperature of the chiplet by 29.17°C and 63.83°C lower than the heat sink system. For the 3D chiplet system, the advantages of microfluidic cooling have a bigger impact than the 2.5D integration.

In the second example, we model a processor-processor integration, a thermally severe case. The hypothetical processor architecture is modeled based on the Intel i7-3960X [20], [21]. Figure 2 shows the floorplan of still square with a side length of 20.8 millimeters generated with ArchFP. Each processor has integrated side by side for the 2.5D chiplet integration, and for the 3D chiplet, one processor is stacked directly on the other processor. We also created a hypothetical power map based on the i7-3960X's published TDP of 130 watts. The power-thermal characteristics are investigated under the heat sinks with convection thermal resistance of 4.0, 2.0, 0.5, and

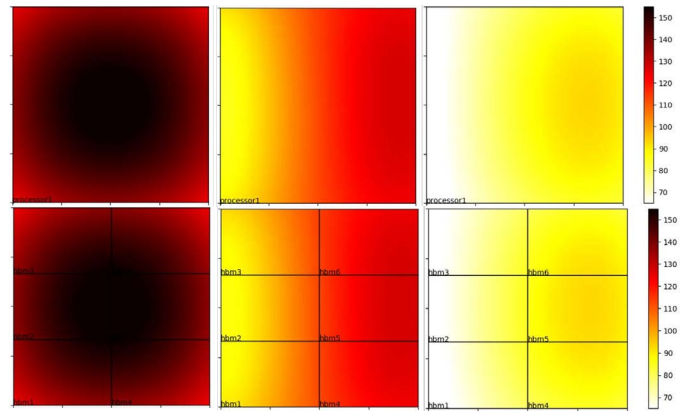


Fig. 8: Temperatures throughout a 3D chiplet with (a) server heat sink, (b) microfluidic cooling, and (c) hybrid of heat sink and microfluidic cooling at power dissipation of 100 watt.

0.2 K/W. And we use the same two configurations for the microfluidic cooling layer with 31 microchannels of a height of 100 μm and a width of 325 μm . The microchannels have a flow rate of 53.5 ml/min under the pressure of a 200 mbar pump supply.

In the second place, we evaluated the thermal characteristics of a processor-processor integration. The hypothetical processor based on i7. The hypothetical power map assumed that six of eight cores are active and consuming 20.5W per core and two cores are disabled for thermal management. The other blocks, IO, L3-cache, and memory controller are consuming 3W, 2W, and 2W, respectively. We normalized the power consumption with a maximum TDP of 130 watts. When the power consumption of the processor is 130W, the maximum temperature of the 2.5D chiplet with high-end and server heat

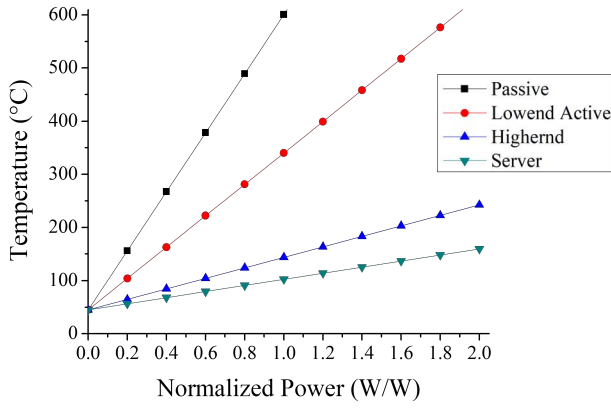


Fig. 9: Temperature of 2.5D high-performance chiplet with heat sinks.

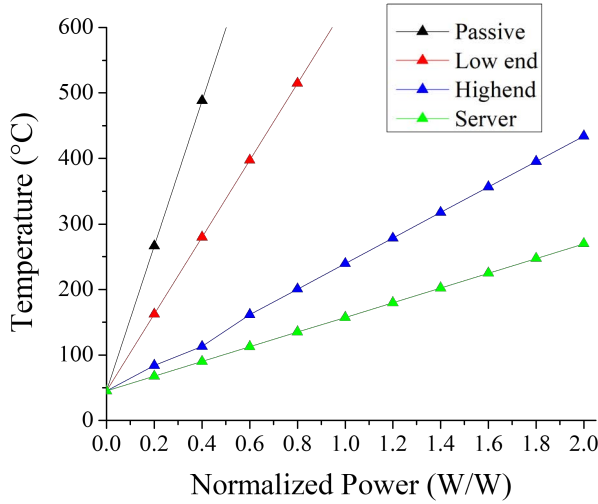


Fig. 10: Temperature of 3D high-performance chiplet with heatsinks.

sinks are 143.79°C and 102.34°C, respectively. The maximum temperatures of processor-processor stacked 3D chiplet at the same condition are 239.58°C and 157.41°C.

High-performance processor-processor 3D chiplet integration exacerbated the inherent thermal problem. For microfluidic cooling, the maximum temperature of 2.5D chiplet decreases 4.56°C lower than the server heat sink. In addition to the microfluidic, with the top surface heat sink, the operating temperature decreases 24.11°C. In the same way, microfluidic cooling decreases the the maximum temperature of the 3D chiplet by 8.24°C and the hybrid cooling decreases the temperature by 48.22°C

IV. DISCUSSIONS

The simulation results with air convection-type heat sinks show that the conventional cooling methods are not suitable for

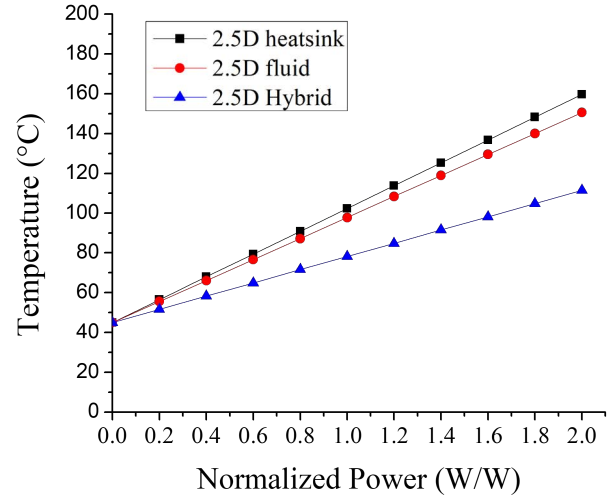


Fig. 11: Temperature comparison of 2.5D high-performance chiplet with server heat sink, microfluidic cooling, and hybrid of heat sink and microfluidic cooling at normalized power dissipation 130W.

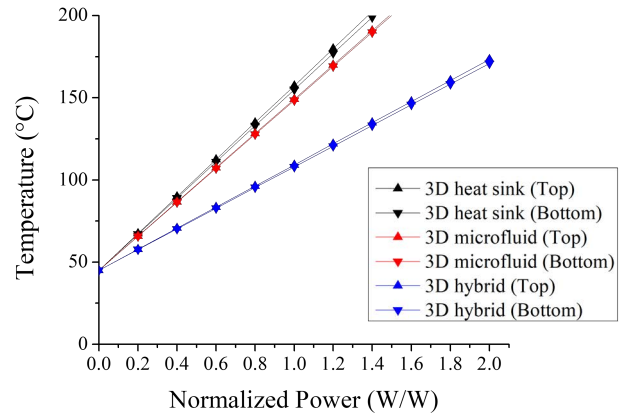


Fig. 12: Temperature comparison of 3D high-performance chiplet with server heat sink, microfluidic cooling, and hybrid of heat sink and microfluidic cooling at normalized power dissipation 130W.

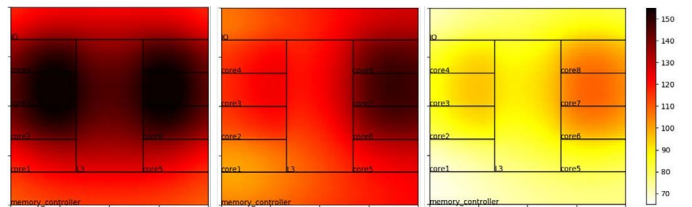


Fig. 13: Temperatures throughout a 3D high-performance chiplet with (a) server heat sink, (b) microfluidic cooling, and (c) hybrid of heat sink and microfluidic cooling.

maintaining the temperature of the current cutting-edge chiplet under the operating temperature range. Although microfluidic is advantageous to heat sinks in cooling behavior, it still lacks the cooling ability to maintain the operating temperature under the limit. Increasing the pump pressure can increase the flow rate, thereby decreasing the temperature. Figure 12 shows the maximum temperature of the 3D chiplet composed of a processor and memories increasing pump pressure. The temperature of the chiplet reduces 82.63°C by solely a microfluidic cooling system with a flow rate of 107.1 ml/min under the pressure of a 400 mbar pump supply. Increasing pump pressure can cause mechanical stress in the device and increase cooling costs. Adding another cooling layer is an alternative solution for the high-performance system. The simulation results shown in Figures 12 and 13, we integrate a conventional heat sink on the top with microchannels at the interposer layer. By substituting the heat sink into the microchannel cooling layer, the temperature reduces to 105.77°C with a flow rate of 106.5 ml/min under the pressure of a 200 mbar pump supply. For the typical 3D chiplet system, microfluidic cooling is a promising thermal management solution. Moreover, for the high-performance 3D chiplet system, a multi-layer microfluidic cooling can be an effective thermal management solution.

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